

Akanksha Ashish Mathpati

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Data Scientist and ML Engineer with hands-on experience building and deploying machine learning systems across NLP, geospatial intelligence, and GenAI. Skilled in developing LLM-powered applications, RAG pipelines, and end-to-end ML workflows that translate complex data into production-ready solutions.

EDUCATION

University of Chicago Oct 2024 - Dec 2025
Master of Science, Applied Data Science GPA: 4.0/4.0
Coursework: Big Data and Cloud Computing, Statistical Models, Machine Learning, Time Series, NLP, Generative AI, ML Ops

Dr. D Y Patil's Ramrao Adik Institute of Technology Oct 2020 - Jun 2024
Bachelor of Technology in Computer Engineering GPA: 9.99/10
Coursework: Database Management Systems, Data Warehousing & Mining, Artificial Intelligence, Big Data Analytics

EXPERIENCE

Blue AI Labs (Nevara) Chicago, IL
Machine Learning Engineer Jun 2025–Aug 2025

- Co-led development of an AI-powered sales coaching platform, transforming transcripts into structured coaching insights with **82% alignment** to expert-coach written reports, providing a **foundation for production deployment**.
- **Built scalable ML workflows (FastAPI + Supabase)** for transcript ingestion, **LLM-based insight extraction**, and XML report generation, reducing manual **review time by 60%** and enabling consistent downstream integration.
- Applied advanced **prompt engineering and evaluation techniques** (multi-call context weighting, schema validation, contradiction checks) to improve reliability and cut output errors by **30%**, strengthening trust in AI-generated insights.

Chief Technology Office, University of Chicago Chicago, IL
Machine Learning Engineer Nov 2024–present

- Implemented features and tested new features in **Phoenix AI, a ChatGPT-like conversational AI system** tailored for university services using **Python, Git, GPT-4o, Docker, Azure OpenAI, Cosmos DB** to improve usability.
- Developed and evaluated a Retrieval-Augmented Generation (**RAG**) model for HR and Compliance teams, improving retrieval accuracy and streamlining application processing using **Python, LangChain, Ollama, and Vertex AI**.
- Optimized and deployed **ML-powered applications**, including LibreChat, by integrating model outputs into existing business intelligence frameworks enhancing accessibility, reliability, scalability, and seamless integration.
- Collaborated with **non-technical stakeholders** across HR, Compliance, and operations to translate **business requirements into ML product features**, ensuring model outputs aligned with real-world workflows.

Here Technologies, University Industry Capstone Chicago, IL
Data Scientist Apr 2025–Dec 2025

- Built an **ML-driven change-detection framework** to identify stale or inconsistent **Points of Interest (POI)** by comparing API outputs against location and operational patterns, enabling **automated detection of map inaccuracies**.
- **Engineered feature-based classification and freshness scoring models** to prioritize high-impact POI updates such as store closures and relocations, **improving signal precision while reducing unnecessary manual review**.
- Designed an **end-to-end ML pipeline** to ingest multi-source location data, **normalize schemas**, perform temporal feature extraction, and generate **reliable freshness signals at enterprise scale** for downstream mapping systems.

PUBLICATIONS and PROJECTS

Research Paper on Dynamics of Cyberbullying Jan 2024
Author, 14th International Conference on Cloud Computing, Data Science and Engineering

- Evaluated multiple **ML classification models** for cyberbullying detection, integrating user engagement signals and **sentiment features** to model content propagation dynamics and **improve detection robustness**.

MedsXplore-Unlock Medicinal Knowledge, (Officially copyrighted by Government of India) Feb 2024
- Built an **OCR + NLP-driven data pipeline** to extract, normalize, and structure medicine information, delivering multilingual drug insights on composition, side effects, and usage through an **AI-powered application**.

SKILLS & PROFICIENCIES

Statistical Analysis & Modeling: Regression, Classification, Time Series Analysis, Feature Engineering, Model Evaluation, Cross-Validation, Error Analysis, A/B testing and hypothesis testing

Programming Languages: Python, R, SQL, PySpark

Machine Learning & NLP: scikit-learn, PyTorch, TensorFlow, XGBoost, Hugging Face, spaCy, Fine-tuning, Transformer Models, BERT

LLMs & GenAI: LangChain, Prompt Engineering, Retrieval-Augmented Generation (RAG), OpenAI API, Chroma DB

Data Engineering & Pipelines: Pandas, NumPy, Spark, ETL Pipelines, Data Modeling, BigQuery

MLOps & Deployment: MLflow, Airflow, FastAPI, Docker, CI/CD, Model Serving, Monitoring, Inference Optimization

Cloud & Visualization: AWS, GCP, Azure OpenAI, Vertex AI, Tableau, Power BI, matplotlib